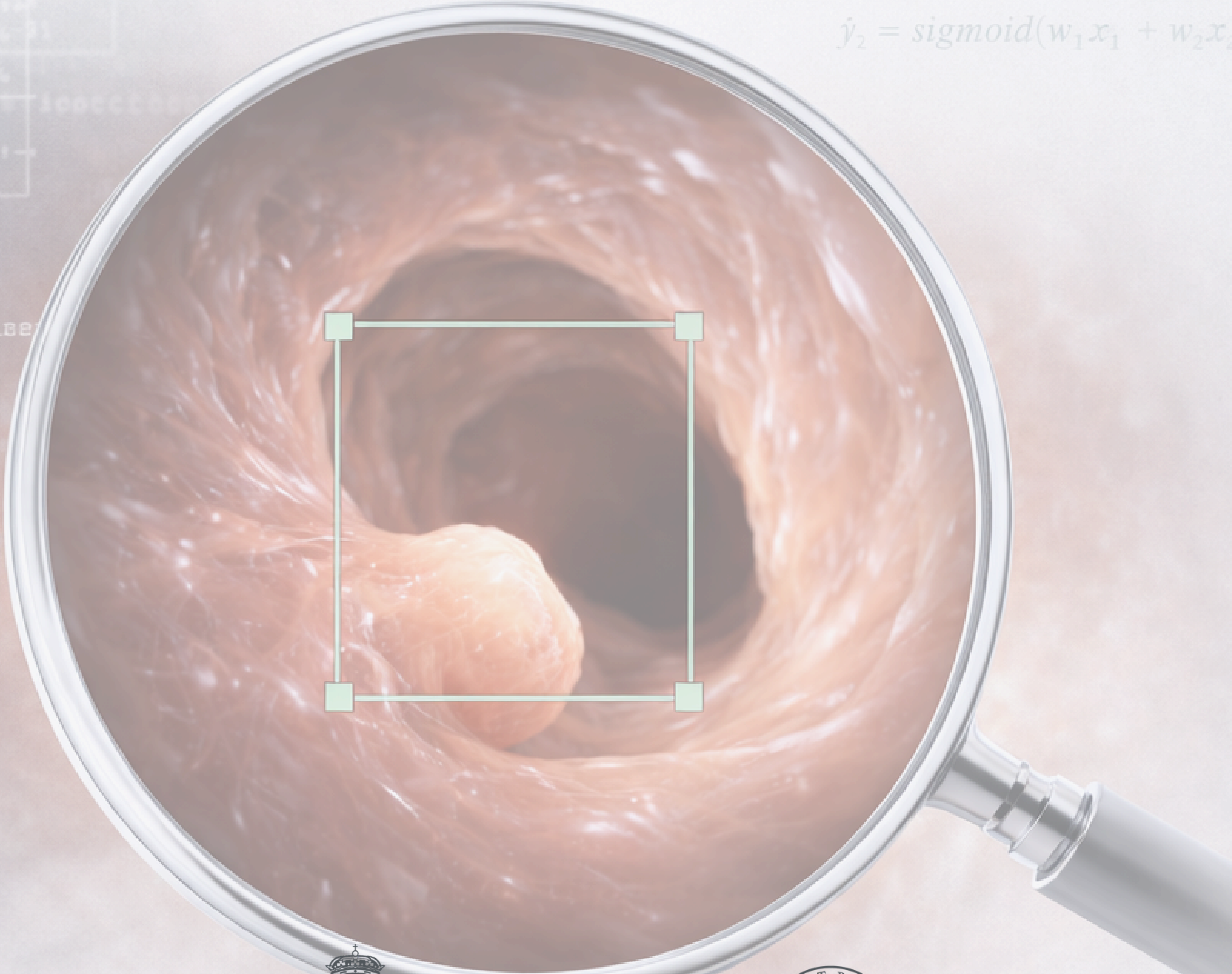
Deep Learning for polyp detection



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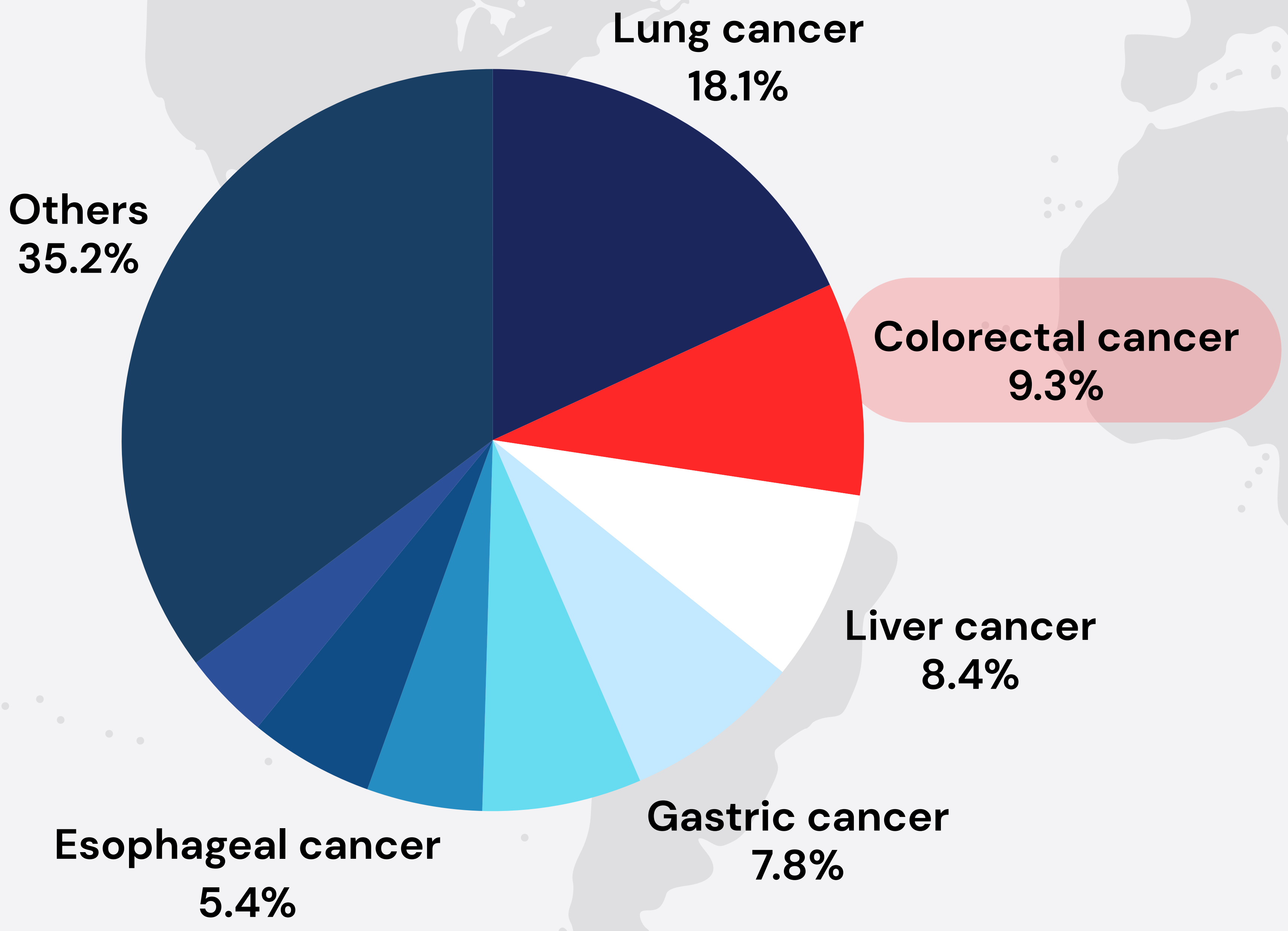
CONTENT

Introduction



[1]

Why current practice is not enough?

- Some lesions can be missed in routine colonoscopy
- Human performance is variable

Why DL has measurable outcome benefits?

- Detect precancerous polyps
- Adenoma Detection Rate (ADR)
- +1% increase in ADR → ~ 5% reduction in fatal interval colorectal cancer risk [2]
- More detected polyps → fewer lesions left to progress
- Impact on stage at diagnosis
- System-level benefits

Objectives

1 Data Loading
Loading and preprocessing of the medical imaging dataset

2 Data Exploration
Exploratory analysis of images and annotations

3 Define models
Selection and configuration of detection architectures

4 Training phase
Model training and optimization

5 Evaluation
Quantitative and qualitative evaluation

Comparative analysis

Data loading

Name of dataset	KVASIR- SEG
Type of images	Colonoscopy images
Total number of images	1000
Image format	JPEG
Image resolution	332×487 to 1920×1072
Type of annotation	Bounding Boxes (+ Seg. Masks)

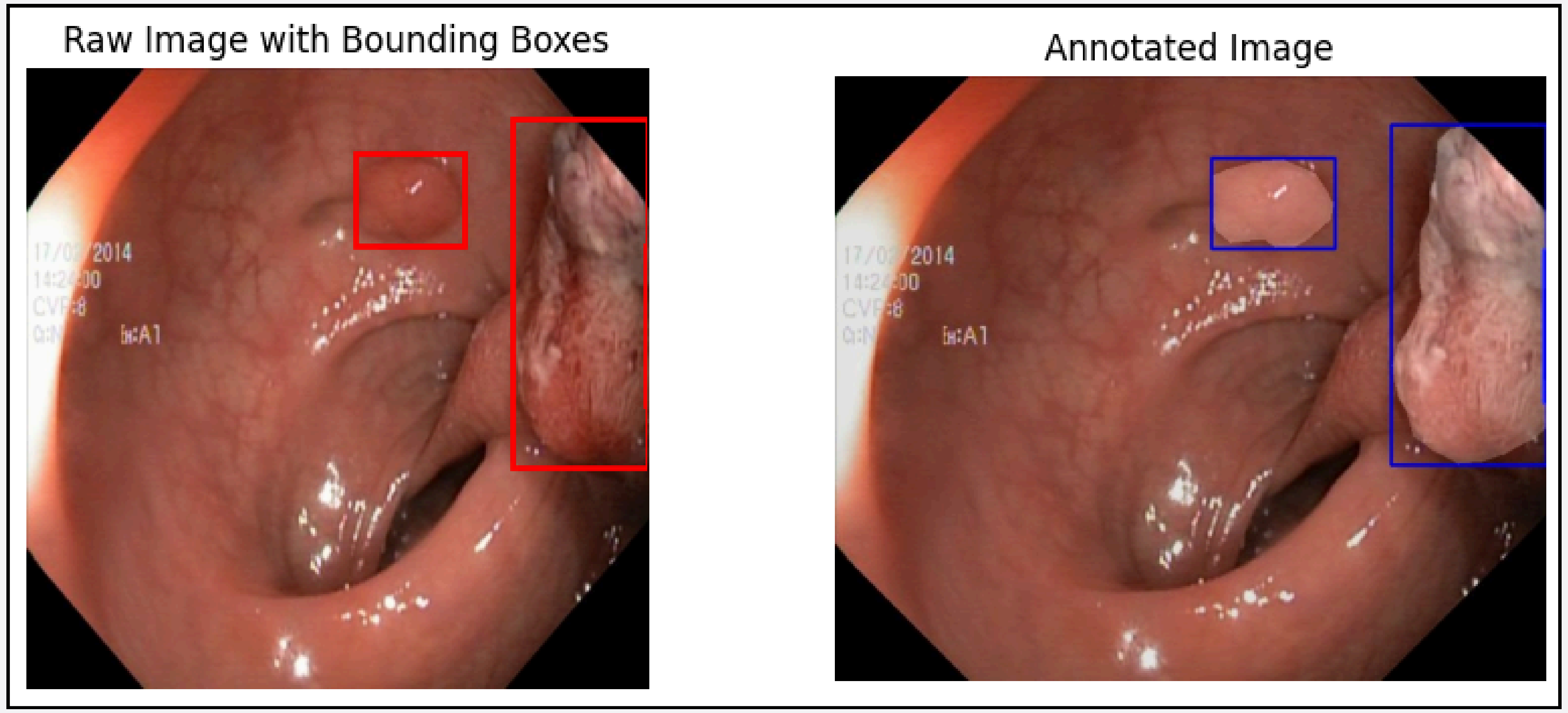


Figure 1. Colonoscopy Image with Ground Truth Bounding Boxes

KVASIR – SEG

- images: original images
- bbox: bounding boxes associated with the original images (.csv)
- annotated images: Colonoscopy images on which the segmentation masks/bounding boxes are defined

Total Bounding boxes = 1071

Invalid Bounding boxes = 0

Data exploration

Positive correlation and cluster in the expected region, but also substantial variability / derivations

Aspect ratio of bounding boxes roughly normally distributed around 1

Size distribution skewed towards small objects; at the same time substantial amount of bigger polyps

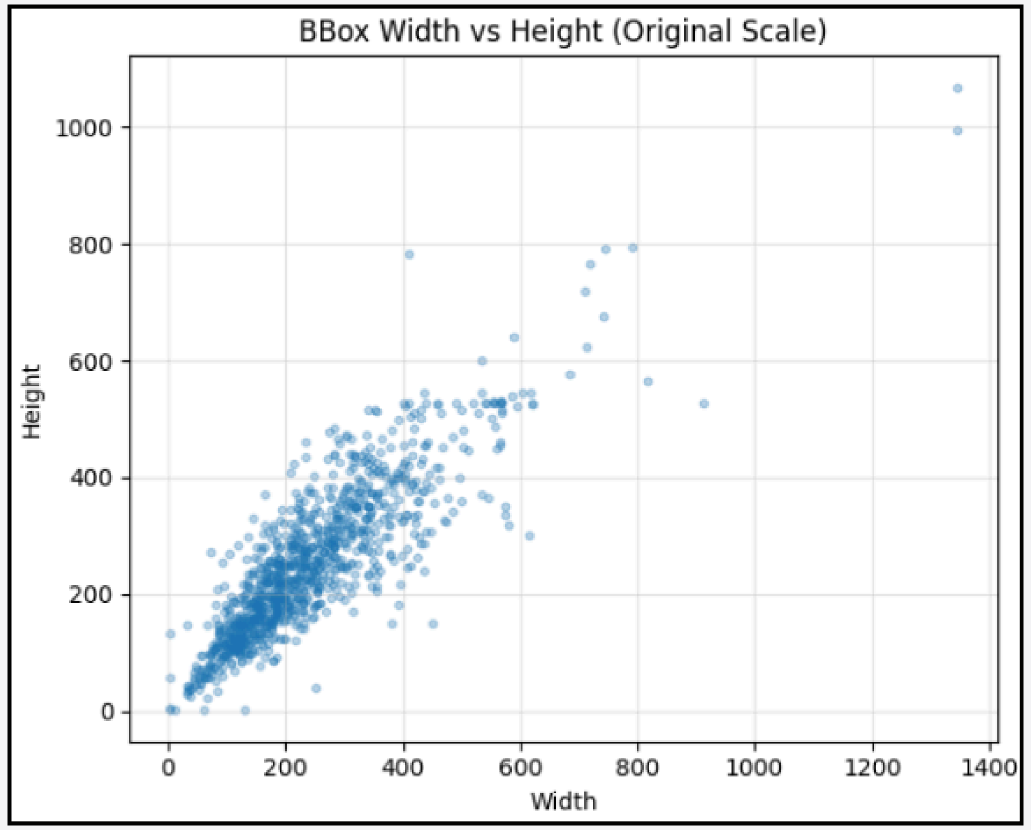


Figure 2. Distribution of Bounding Box Dimensions (Width vs Height)

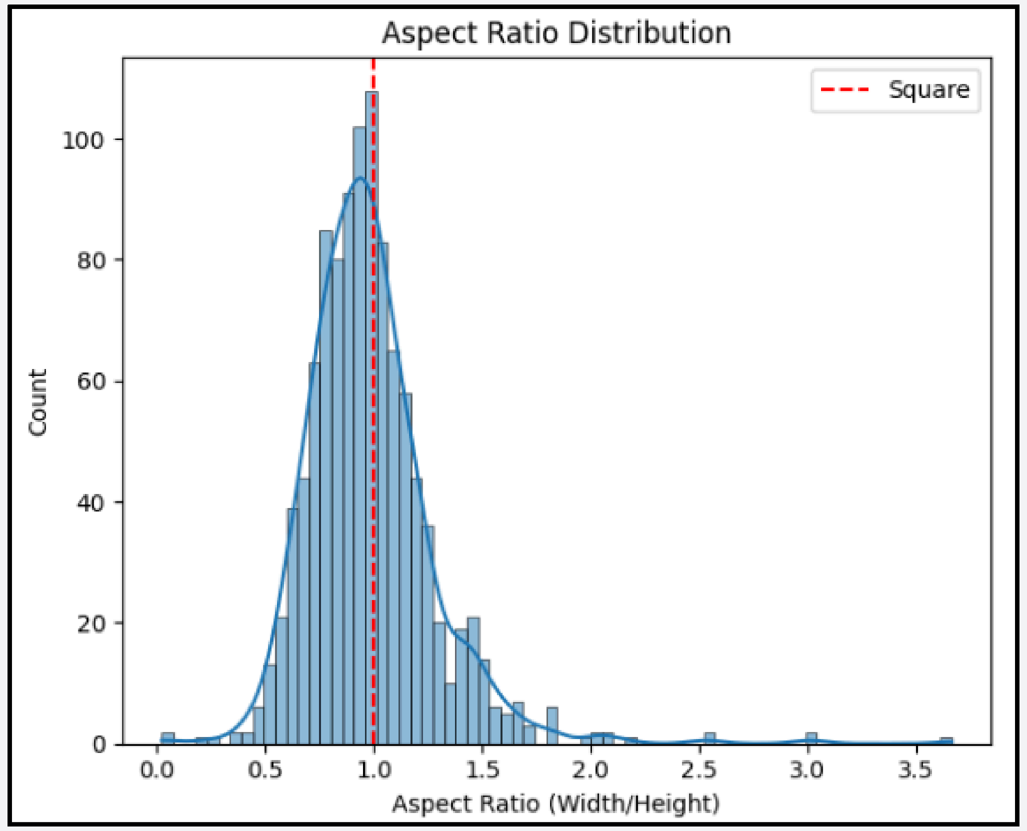


Figure 3. Aspect Ratio Distribution of Bounding Boxes

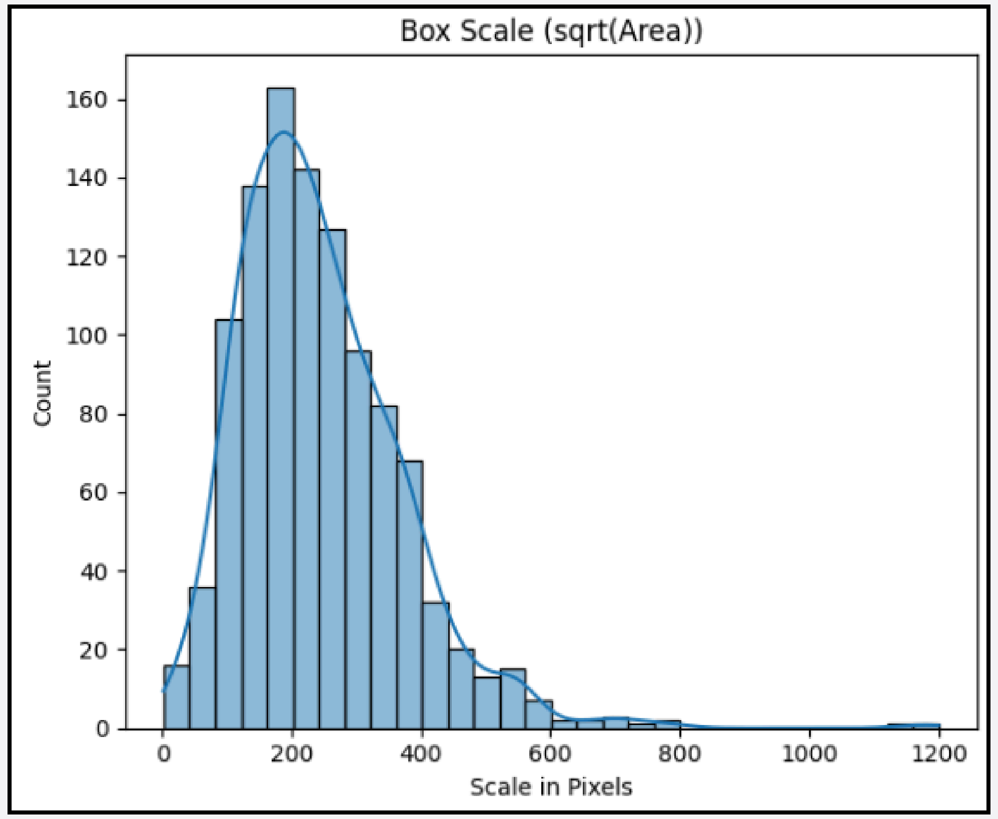
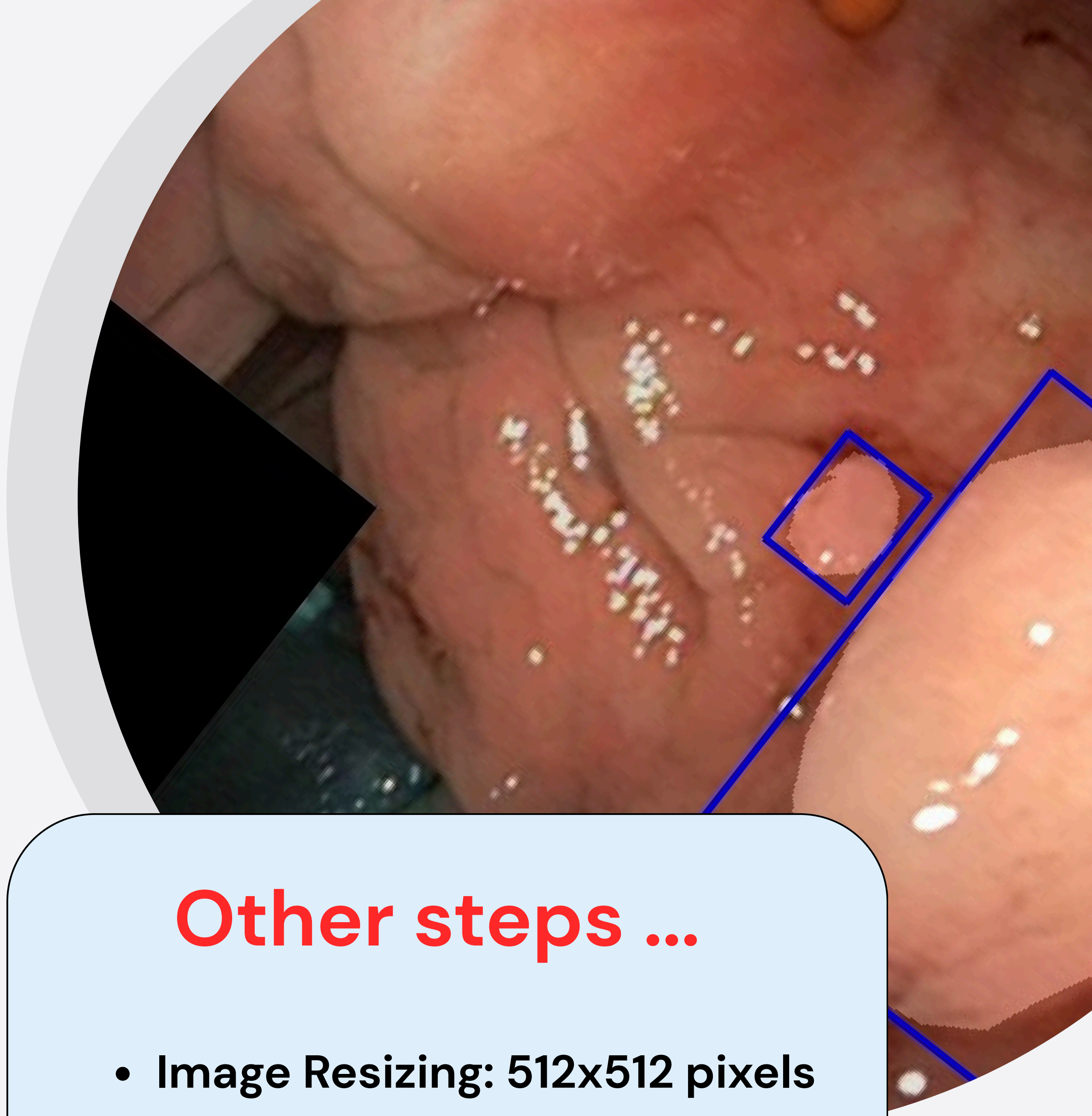


Figure 4. Bounding Box Scale Distribution

Data Preprocessing



Data Normalization

RGB Image

↓

Floating point tensors scale to [0,1] range by dividing raw pixel intensities by 255.

Data Augmentation

- Horizontal/vertical flips
- Small rotations
- Scale and random resize/crop
- Brightness/contrast perturbations

Other steps ...

- Image Resizing: 512x512 pixels
- CLAHE Preprocessing to enhance local contrast.

Model selection and design

Two-stage detectors

Faster R-CNN + FPN *with...*

ResNet-50 backbone
(CNN)
+
Quantile based
anchor proposals

Model 2

Swin backbone
(hierarchical transformer)
+
K-Means clustering based
anchor proposals

Model 4

One-stage detectors

YOLO

- Predicts boxes and classes in a single pass
- usually faster inference
- sometimes lower accuracy on small objects
- Serves as SOTA baseline

Model 3

Model Architecture

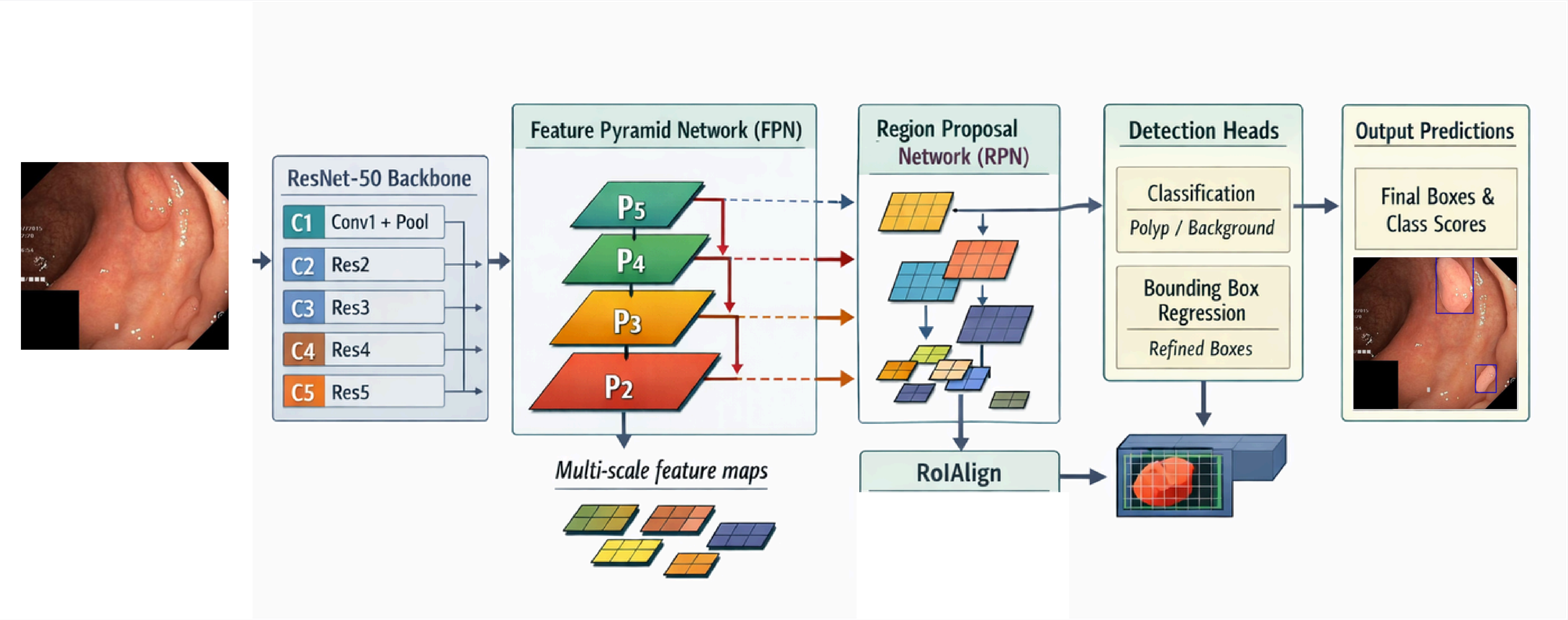


Figure 5. Architecture for RCNN ResNet-50 + FPN

Backbone: ResNet-50/Swin (feature extractor)

Feature Pyramid Network (FPN)

Region Proposal Network (RPN)

RoIAlign / RoI pooling

Detection heads

Region Proposal Network

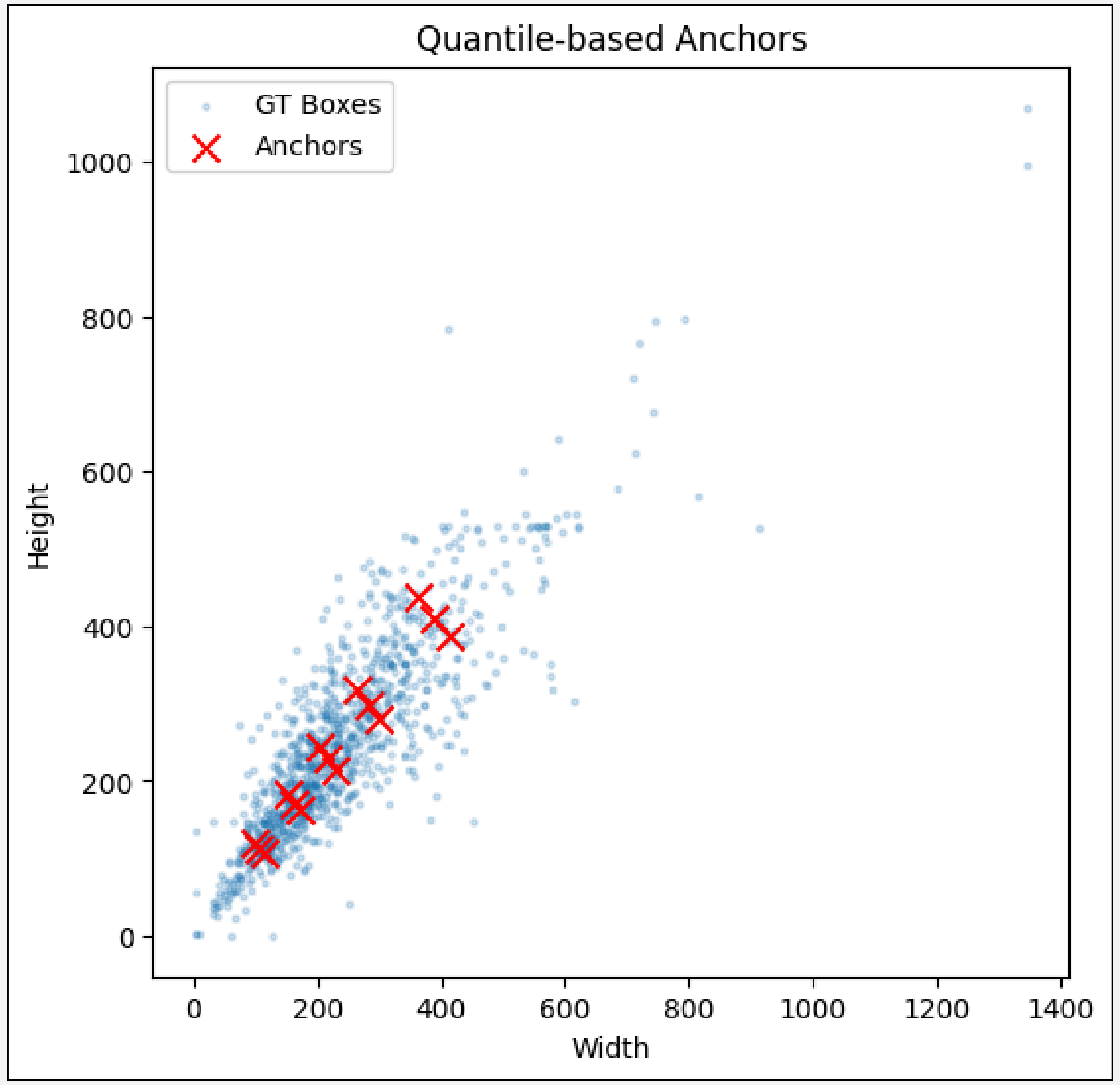


Figure 6. Anchor proposals based on quantiles of aspect ratio and sizes (training set only)

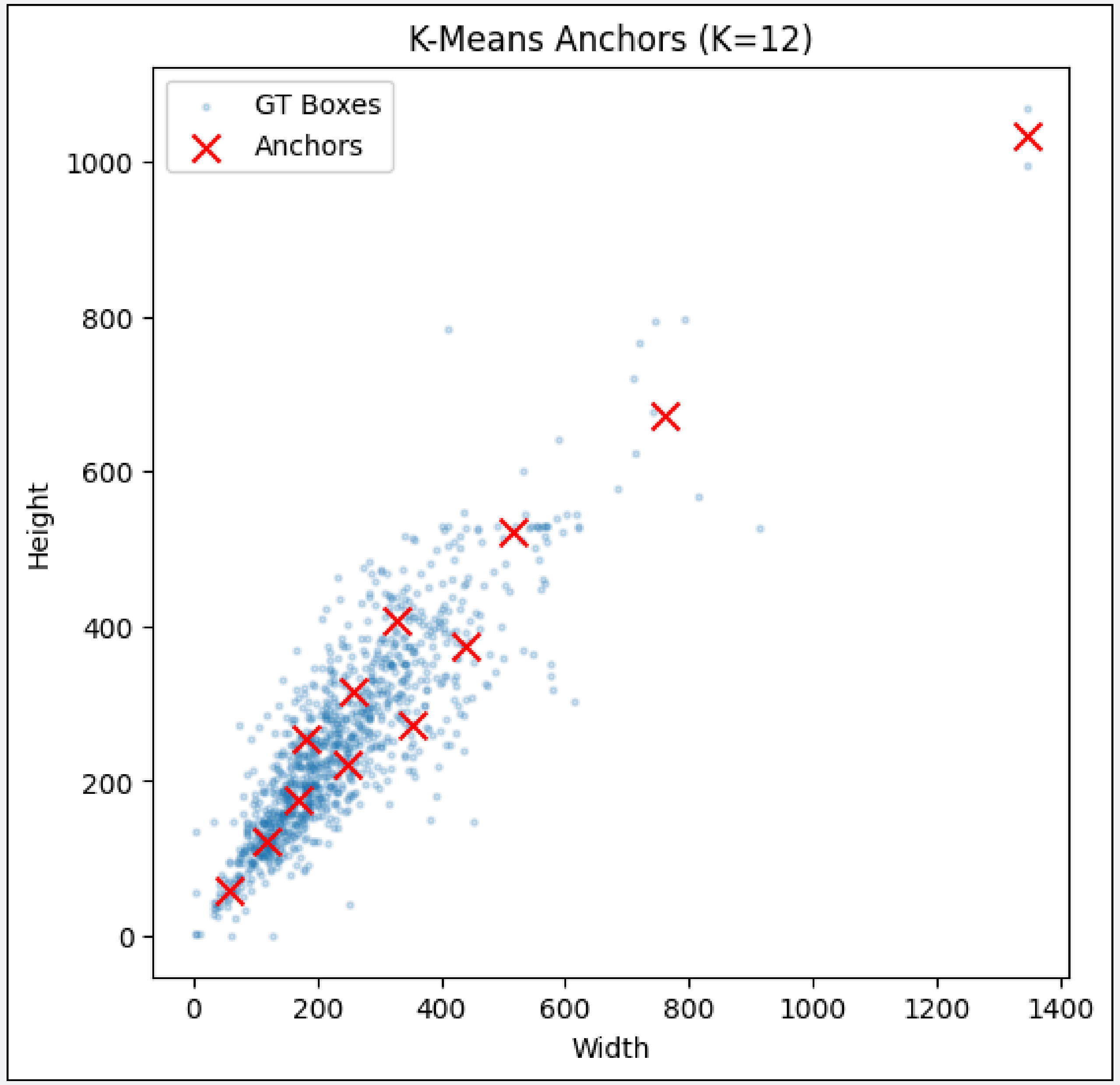
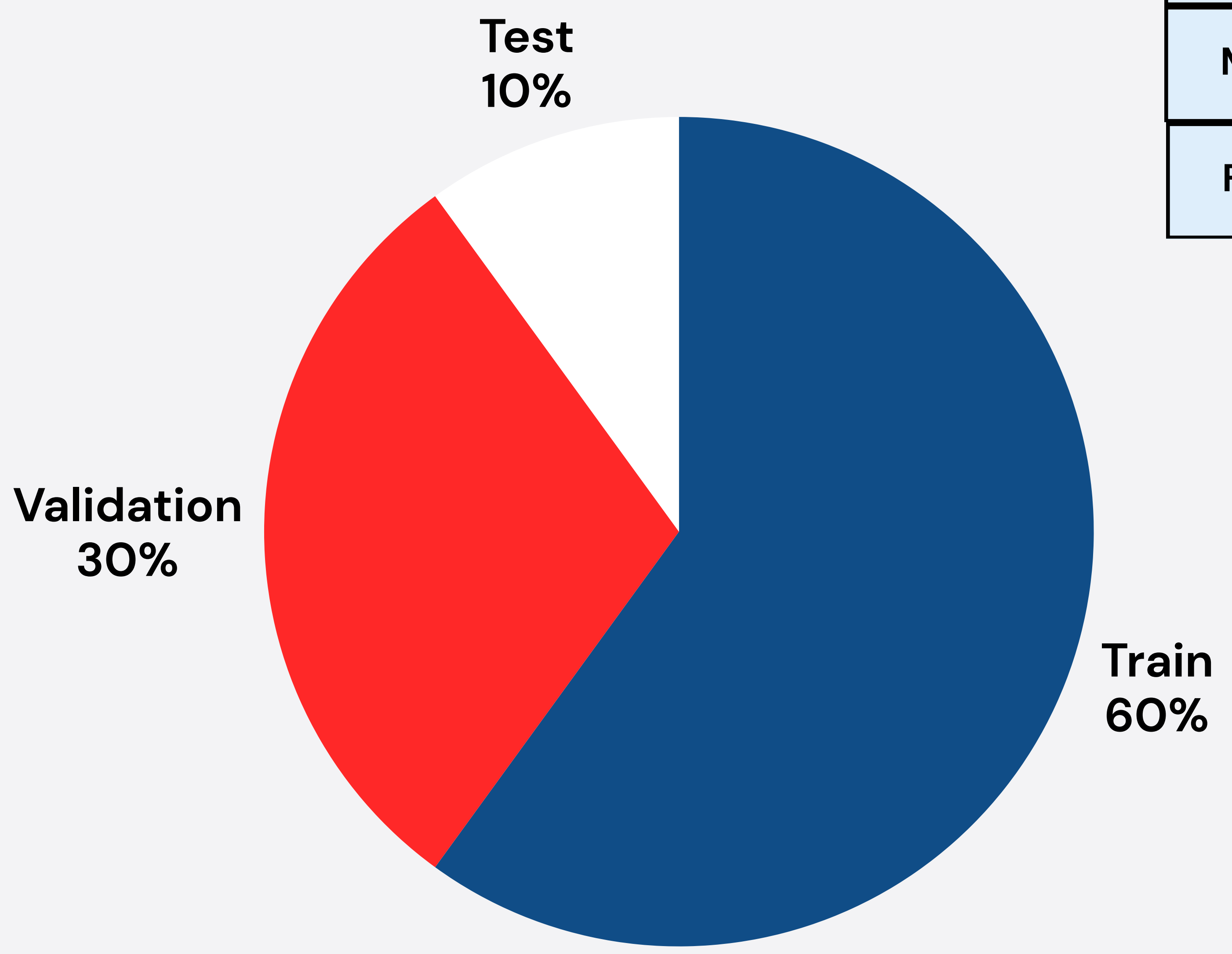


Figure 7. Box Proposals based on K-Means clustering on width and height (training set only)

Training



Parameters	Faster RCNN- FPN with ResNet backbone	Swin-based Faster RCNN -FPN
Optimizer	SGD	Adam
Learning rate	0.002	1e-4
Weight decay	5e-4	1e-4
Momentum	0.9	
Frozen Layers	Conv1 + Bn1 + Layer1	Stem + Stage1

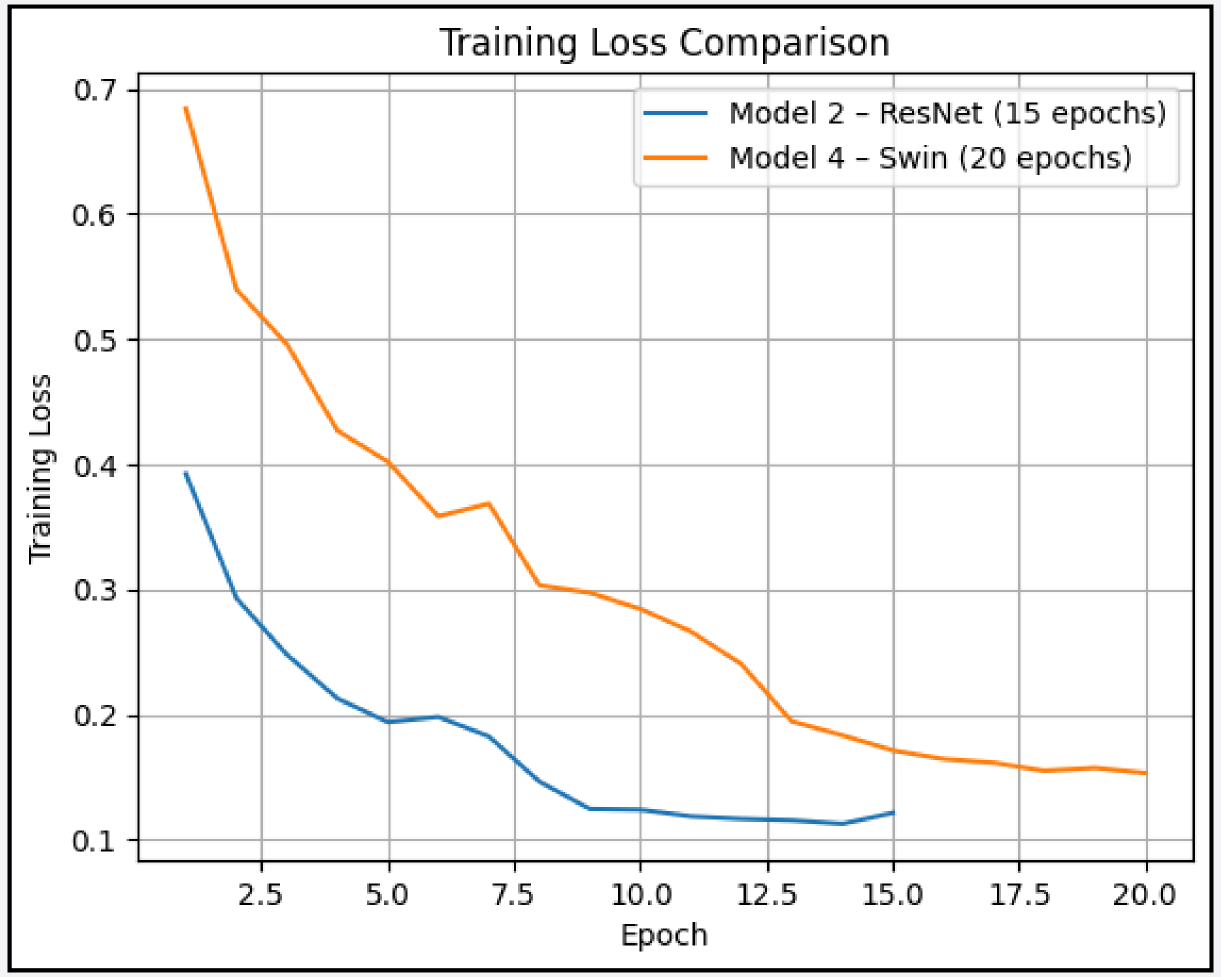


Figure 8. Training loss for ResNet50-FPN and Swin-based ResNet

Evaluation: mAP@50

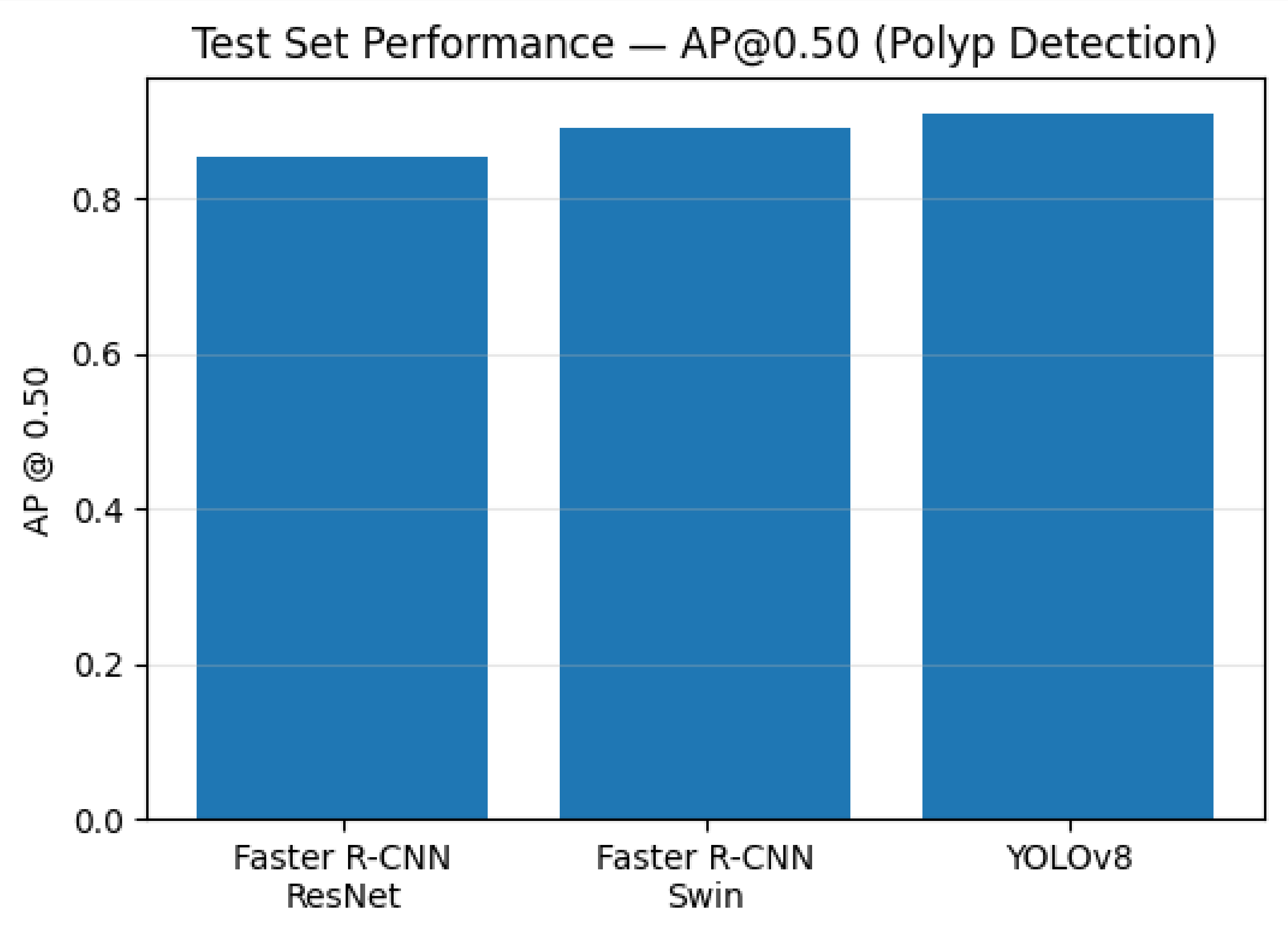


Figure 9. Average Precision with IoU Threshold 0.50

1. **YOLOv8**
2. **Faster R-CNN (Swin)**
3. **Faster R-CNN (ResNet)**

Since AP@0.50 mainly rewards finding the object even if the box is not perfectly tight, this suggests that all models learned robust “objectness” features for polyp presence and produce reliable detections.

Evaluation: mAP@75

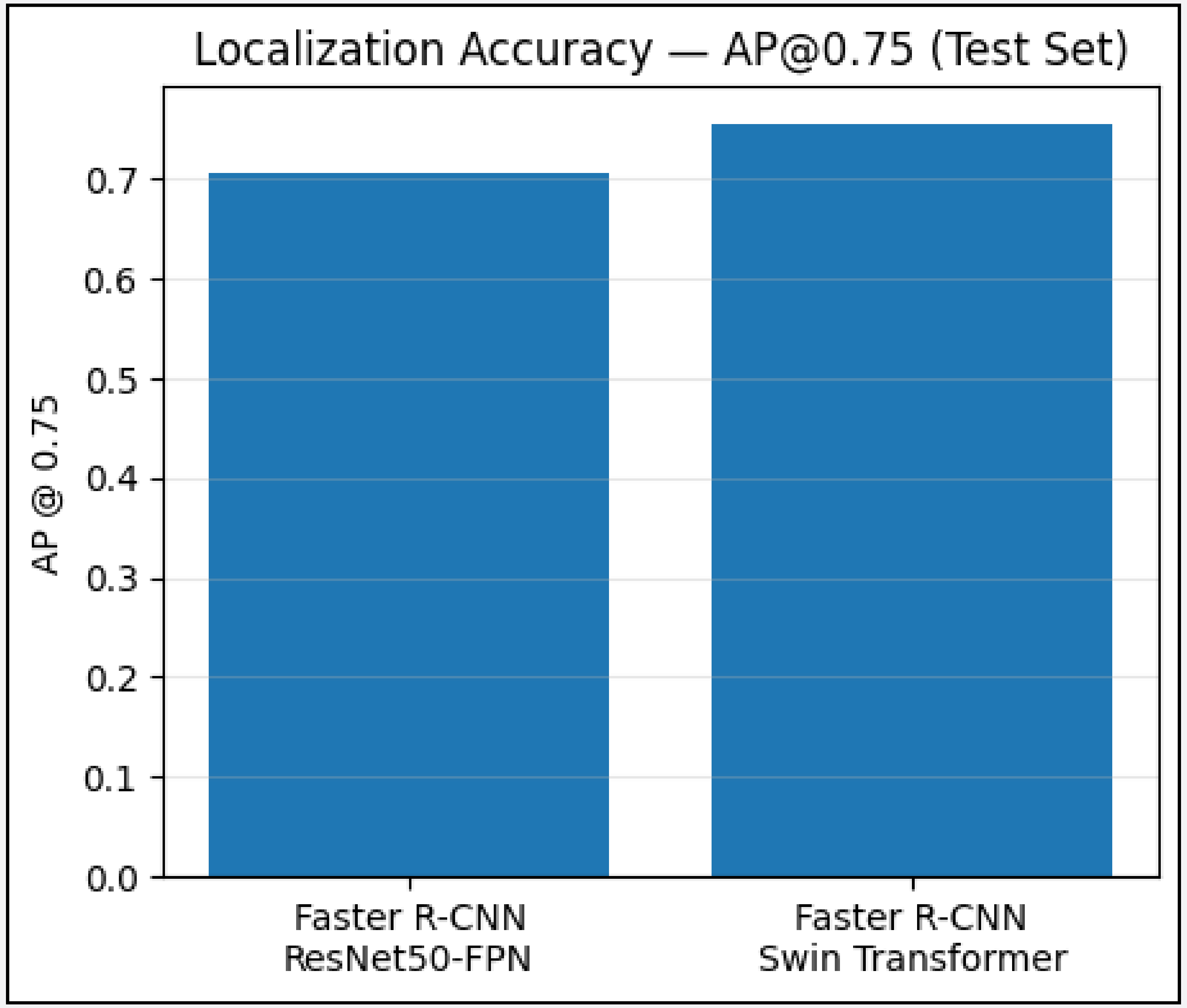


Figure 10. Average Precision with IoU Threshold 0.75

- Swin backbone is helping the detector produce more accurately placed/tighter boxes
- Swin-style backbones tend to capture broader contextual information while still preserving multi-scale structure.

Evaluation: Precision/Mean-IoU/Recall/F1-Score

Metrics	ResNet50 - FPN	Swin-based	YOLO
Precision IoU>0.5	0.449438	0.5989010	0.903808
Mean IoU, IoU>0.5	0.863818	0.821924	0.896787
Recall IoU>0.5	0.936937	0.918919	0.873874
F1-Score IoU>0.5	0.607477	0.725174	0.888589

Precision:

- **Swin backbone improved precision** over ResNet backbone. Least false positives in YOLO

Mean IoU:

- Reflects box tightness conditioned on successful detection and is complementary to AP@0.75, which also accounts for confidence ranking, duplicates and unmatched predictions. **ResNet's** correct detections have **more precise bounding boxes than Swin's**.

Recall:

- **Faster R-CNN with ResNet** achieved the highest recall indicating the **lowest false-negative** rate among the compared models. Although its precision is lower (more false positives), this behavior can be acceptable, or even preferable, in medical screening settings.

F1 - Score:

- Highlights the overall operating-point trade-off between false positives and false negatives

Evaluation:

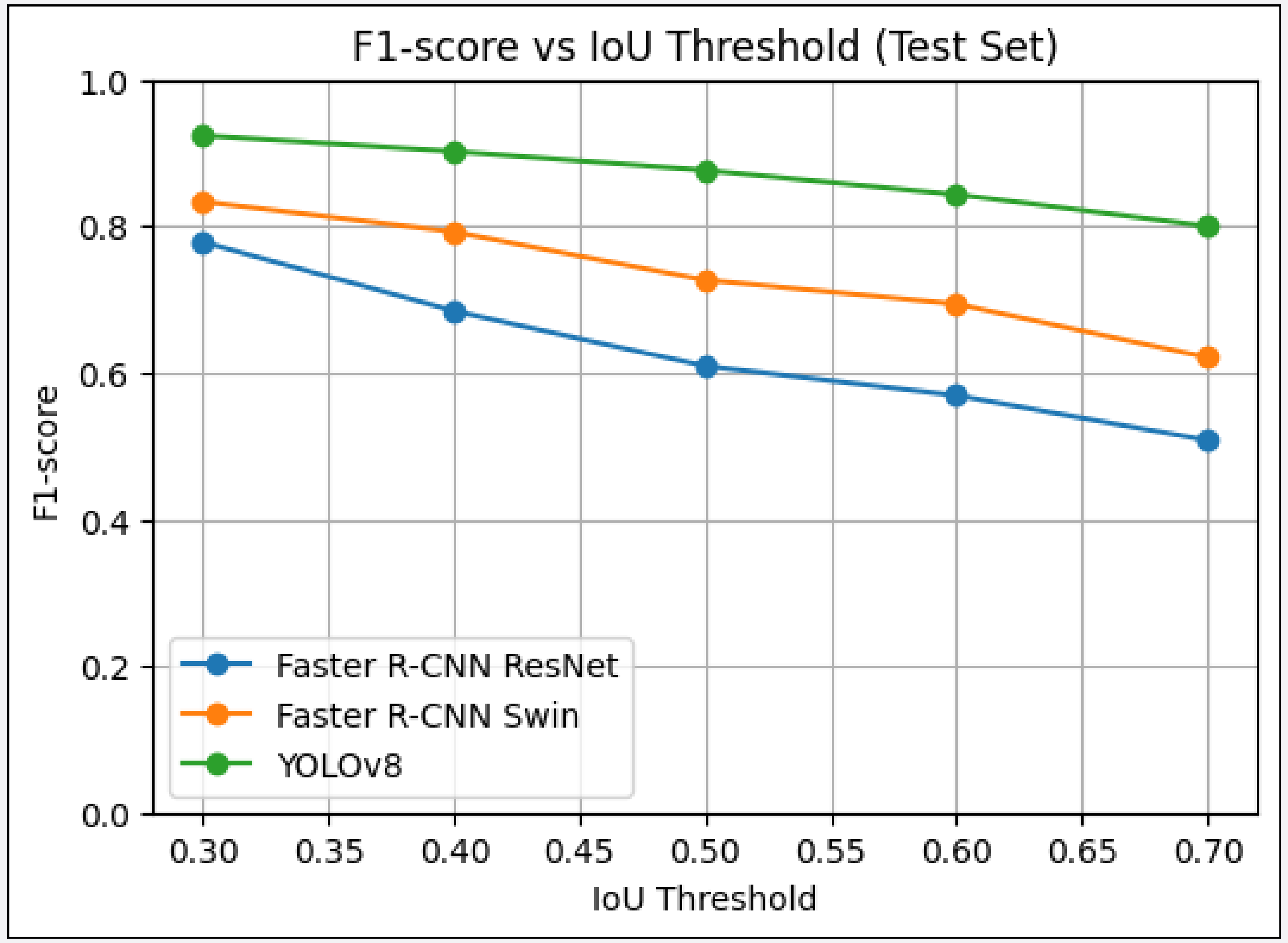


Figure 11. F1 -Score vs IoU Threshold

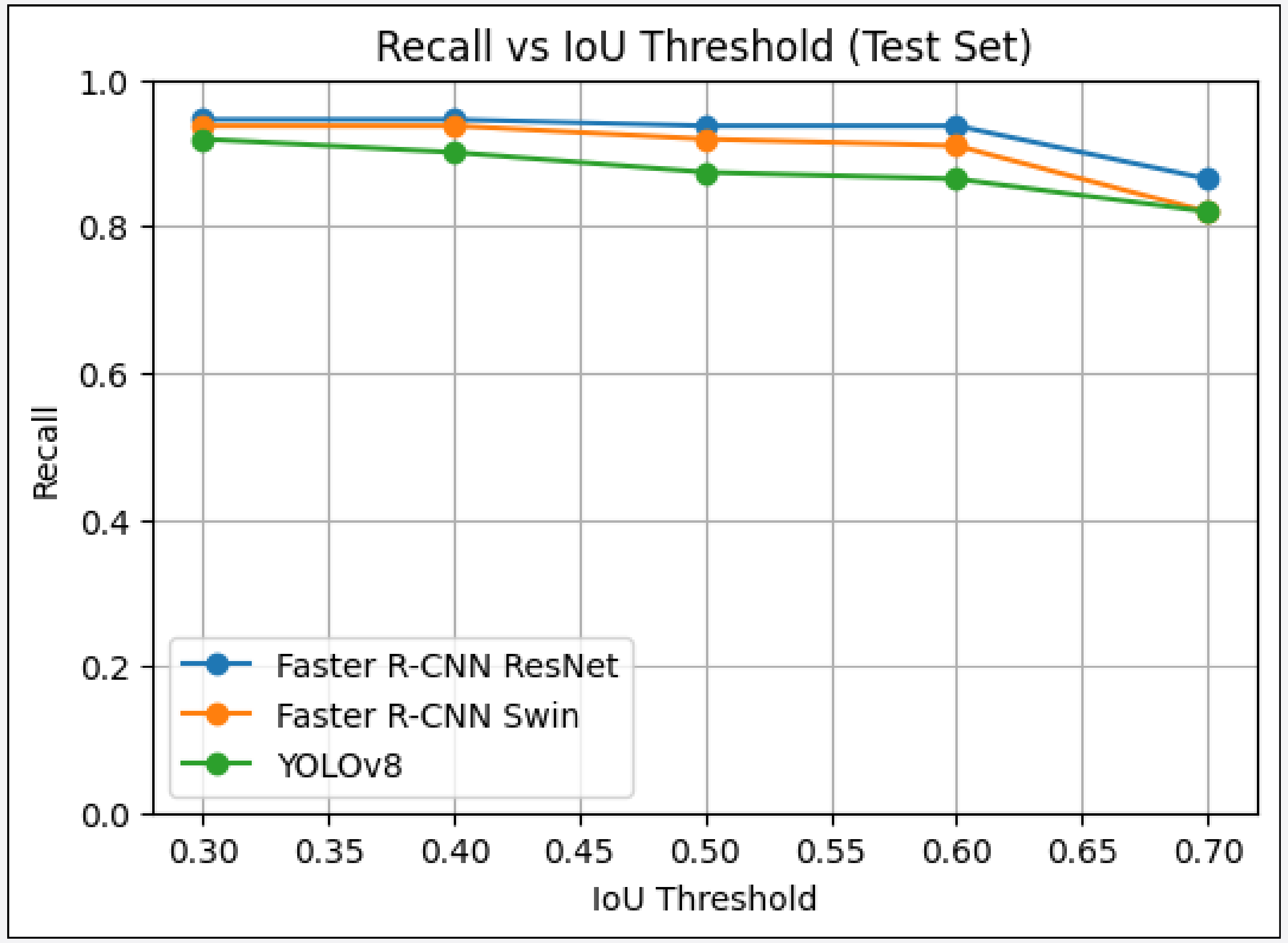


Figure 12. Recall vs IoU Threshold

YOLOv8 achieves highest F1 across thresholds and shows the smallest degradation

most robust overall operating behavior under increasingly strict localization. .

Across IoU thresholds from 0.30 to 0.60, both Faster R-CNN variants maintain very high recall with only a mild decrease, indicating robust detection under increasingly strict matching.

Conclusions:

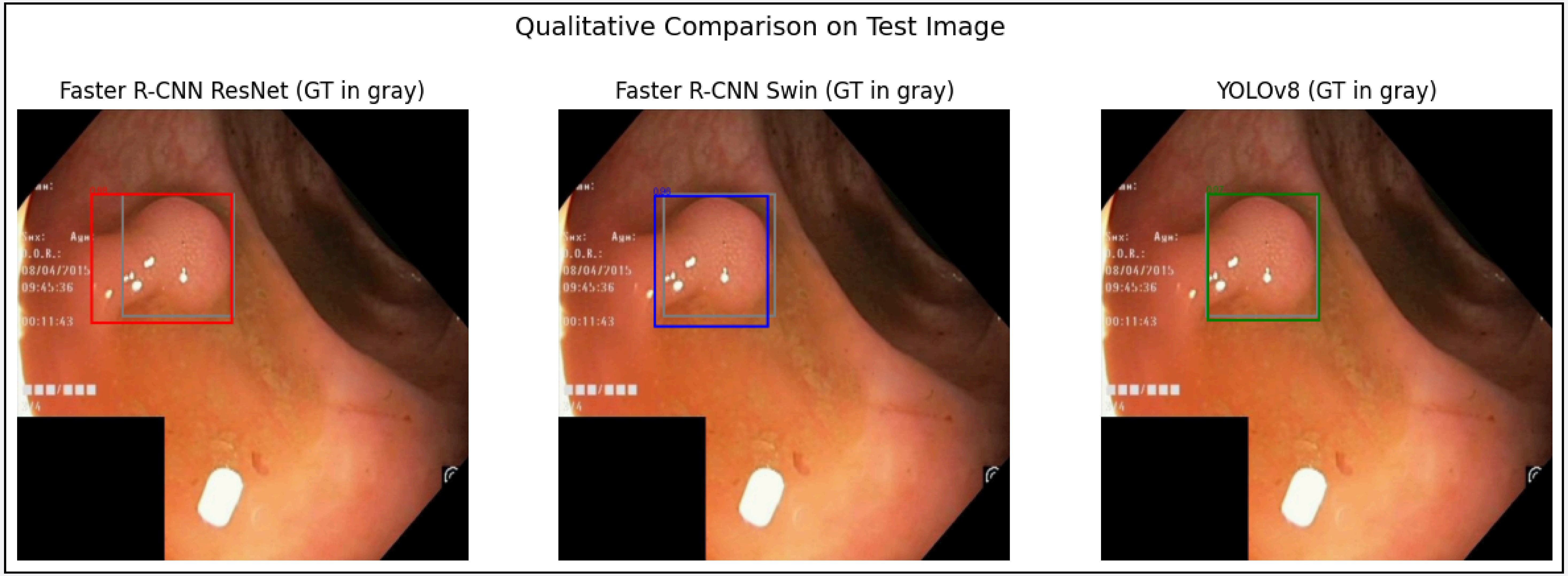


Figure 13. Qualitative comparison of three models

The Swin-based Faster R-CNN with k-means–derived box proposals achieved stronger overall detection performance, while the ResNet-based model using quantile-based proposals favored higher recall and slightly tighter localization on correct detections.

When compared to YOLOv8 as a reference baseline, both Faster R-CNN variants remain competitive at moderate IoU thresholds, though YOLOv8 achieves stronger overall performance under stricter localization criteria.

Bibliography:

[1] World Health Organization. International Agency for Research on Cancer. (n.d.). International Agency for Research on Cancer (IARC).

[2] Corley, D. A., Jensen, C. D., Marks, A. R., Zhao, W. K., Lee, J. K., Doubeni, C. A., … Quesenberry, C. P. (2014). Adenoma detection rate and risk of colorectal cancer and death. New England Journal of Medicine, 370(14), 1298–1306